

Student Project Proposal 2026:

DFT Study of Transition Metals and Their Oxides

1. Assigned Systems

Each student will study a pair consisting of a transition metal, **M**, and its corresponding monoxide, **MO**.

Student	Metal, M	Oxide, MO	Key Physics
Filip	bcc Fe	FeO, rocksalt	Strong ferromagnetic metal → antiferromagnetic correlated oxide
Bartosz	hcp Co	CoO, rocksalt	Itinerant ferromagnetic metal → localized antiferromagnetic oxide
Stanisław	fcc Ni	NiO, rocksalt	Weak ferromagnetic metal → antiferromagnetic Mott / charge-transfer insulator
Mathew	bcc Mn, model system	MnO, rocksalt	Complex magnetism → strong local-moment antiferromagnetic oxide

2. Project Tasks

A. Convergence Tests

For both the metal **M** and the oxide **MO**, perform convergence tests with respect to:

- Plane-wave cutoff energy, **ENCUT**
- k-point mesh

Assume magnetic state for M (FM order) and MO (AFM order)

For each system, plot:

- Total energy vs ENCUT
- Total energy vs k-point mesh (number of IBKZ points)

Choose the final computational parameters based on a clear convergence criterion, for example:

Total energy convergence better than 1 meV/atom.

Include a summary table:

System	ENCUT, eV	k-point mesh	Energy convergence
M			
MO			

B. Magnetic Ground State and Structural Relaxation

Before structural relaxation, determine the magnetic ground state of the M and MO structures.

B1. Metals, M

Compare at least:

- Non-magnetic, NM
- Ferromagnetic, FM

For systems where relevant, additional magnetic configurations may also be considered.

Report:

- Total energy difference between magnetic states

- Magnetic moment per atom
- Final relaxed lattice parameters
- Residual forces
- Stress tensor

B2. Oxides, MO

For the oxides, compare at least:

- Ferromagnetic, FM
- Antiferromagnetic, AFM

Perform calculations using:

- Standard PBE, without U
- DFT+U, using reasonable values of U, for example:
U = 3eV, with J = 1 eV - The choice of U and J must be justified using literature values.

Analyze:

- Stability of AFM vs FM states
- Local and total magnetic moments
- Effect of U on lattice parameters, magnetic moments
- Comparison with experimental and theoretical literature values

For each magnetic state, include:

- Total energy
- Magnetic moments
- Final lattice parameters
- Residual forces
- Stress tensor

Also include visualizations of the initial and relaxed structures.

For the lowest-energy relaxed magnetic state (magnetic ground state) calculate cohesive energy, which is defined as following:

- for a metal supercell containing N_M metal atoms:

$$E_{\text{coh}}(M) = \frac{N_M E_{\text{atom}}(M) - E_{\text{tot}}(M_{\text{bulk}})}{N_M}$$

- for an oxide supercell containing N_M metal atoms and N_O oxygen atoms:

$$E_{\text{coh}}(MO) = \frac{N_M E_{\text{atom}}(M) + N_O E_{\text{atom}}(O) - E_{\text{tot}}(M_x O_y)}{N_M + N_O}$$

where $E_{\text{coh}}(M)$ is cohesive energy per metal atom, $E_{\text{atom}}(M)$ total energy of an isolated metal atom, $E_{\text{tot}}(M_{\text{bulk}})$ total energy of the relaxed metal bulk cell and N_M number of metal atoms in the cell.

Example summary table:

System	Method	Magnetic ordering	a, Å	b, Å	c, Å	α , °	β , °	γ , °	Max force, eV/Å	Stress / pressure	E_total [eV] / cohesive energy [eV]
M	PBE	NM									
M	PBE	FM									
Experimental values											
MO	PBE	FM									
MO	PBE	AFM									
MO	DFT+U (U=, J=)	FM									
MO	DFT+U (U=, J=)	AFM									
Experimental values											

C. Electronic Structure

For the relaxed structures and magnetic ground state, compute:

- Band structure
- Total density of states, DOS
- Spin-resolved DOS
- Projected density of states, PDOS
- Projected band structure showing orbital character

For PDOS, include:

- Metal atom: s, p, and d orbitals
- Oxygen atom: s and p orbitals

Analyze:

- Metallic vs insulating behavior
- Band gap value and type, if present
- Orbital contributions near the Fermi level
- Differences between the metal M and the oxide MO
- Effect of DFT+U on the electronic structure of MO

Include a summary table:

System	Metallic / Insulating	Band gap, eV	Dominant orbitals at E_F	Magnetic moment, μ_B
M				
MO				

D. Phonon Calculations

For both **M** and **MO**, compute:

- Phonon frequencies at the Γ -point
- Phonon dispersion relation (plot it)
- Phonon density of states, PhDOS (plot it)

Identify:

- Imaginary modes, if any and dynamical stability

Include a summary table:

System	Mode	Frequency at Γ	Imaginary?	Dynamically stable?
M				
MO				

E. Mechanical Properties

Compute elastic constants using either the stress–strain method or the energy–strain method.

For cubic systems, report relevant nonzero components of elastic tensor (e.g. C_{11} , C_{12} , C_{44})

Then derive:

- Bulk modulus
- Young’s modulus
- Poisson’s ratio

Include a summary table:

System	C_{11}	C_{12}	C_{44}	Bulk modulus	Young’s modulus	Poisson’s ratio
M						
MO						

3. Report Requirements

The final report should include the following sections.

1. Introduction

Include:

- Short background on transition metals and transition-metal monoxides
- Magnetic behavior of metals and oxides
- Motivation for comparing metallic M systems with oxide MO systems
- Brief discussion of correlation effects and the need for DFT+U in transition-metal oxides

2. Computational Details

Include:

- DFT code used, for example VASP
- Exchange-correlation functional, for example PBE
- Pseudopotentials / PAW potentials
- Plane-wave cutoff energy
- k-point meshes
- Convergence criteria for electronic and ionic relaxation
- Magnetic configurations considered
- DFT+U parameters and justification
- Phonon calculation method
- Elastic-constant calculation method

3. Results and Discussion

Suggested subsections:

3.1 Convergence Tests

- ENCUT convergence
- k-point convergence
- Final parameter choices

3.2 Structural and Magnetic Properties

- Initial and relaxed structures
- Optimized lattice parameters
- Magnetic ground state
- Magnetic moments
- Comparison with literature and discussion of possible discrepancies

3.3 Electronic Structure

- Band structures
- DOS and PDOS
- Band gap values
- Orbital contributions near the Fermi level
- Comparison between M and MO

3.4 Phonon Properties

- Γ -point frequencies
- Phonon dispersion
- Phonon DOS
- Dynamical stability

3.6 Mechanical Properties

- Elastic constants
- Bulk modulus
- Young's modulus
- Poisson's ratio
- Mechanical stability discussion

4. Conclusions should summarize:

- Final relaxed structures
- Magnetic ground states
- Metallic or insulating character
- Effect of oxidation from M to MO
- Effect of DFT+U
- Dynamical and mechanical stability
- Main comparison with literature

4. Required Figures:

The report should include:

1. Initial and final structures
2. ENCUT convergence plot
3. k-point convergence plot
4. Band structure plots

5. Total DOS and spin-resolved DOS
6. PDOS plots
7. Phonon dispersion and Phonon DOS plots

All figures should be clearly labeled and discussed in the text. The report should explicitly compare:

- Metal vs oxide structure
- Magnetic moment changes
- Metallic vs insulating behavior
- Role of oxygen p states
- Localization of metal d states
- Effect of electronic correlations in MO
- Differences between PBE and DFT+U

This comparison should be part of the discussion, not only presented in tables.

5. Submission Requirements

Each student should create the Overleaf project:

1. Final report in PDF format
2. All input files and relevant output files should be arranged in folder structure and added to the project

6. Grading Criteria

Completed Tasks	Final Grade
A + B	3
A + B + C	4
A + B + C + D	5
A + B + C + D + E	5+ / Excellent